Demonstration of Integrated AlScN Photonic Devices on 8-Inch Silicon Substrate

Zhenyu Li, Kewei Bian, Xinyi Chen, Xingyan Zhao, Yang Qiu, Yuan Dong, Qize Zhong, Shaonan Zheng, and Ting Hu

***Abstract*—In combination with higher piezoelectric effect and larger optical second-order susceptibility ($\chi^{(2)}$) compared with aluminum nitride (AlN), scandium-doped aluminum nitride (AlScN) platform is considered as a new platform for future integrated photonics. Hence, further studies are necessary to unlock the potential of the AlScN photonic platform and achieve integrated photonic devices with better performance. In this work, we fabricated the highly c-axis-orientated and low-surface-roughness AlScN thin films by reactive magnetron sputtering with 9.6% and 20% Sc concentrations. Then, we fabricated essential photonic devices including directional coupler (DC), Mach-Zehnder interferometer (MZI), and micro-ring resonator (MRR) based on the AlScN photonic platform. The measured extinction ratios (ERs) of MZIs are 22 dB for $Al_{0.904}Sc_{0.096}N$ and 24 dB for $Al_{0.8}Sc_{0.2}N$. The maximum quality (Q) factors of MRRs are 3.2×10^4 for $Al_{0.904}Sc_{0.096}N$ and 2.2×10^4 for $Al_{0.8}Sc_{0.2}N$, while the waveguide bending losses are about 6.1 dB/cm for $Al_{0.904}Sc_{0.096}N$ and 10.1 dB/cm for $Al_{0.8}Sc_{0.2}N$, respectively. These results pave the way towards low-loss integrated AlScN photonic applications such as acousto-optic modulation, nonlinear optical generation.**

***Index Terms*—Directional coupler, integrated photonics, Mach-Zehnder interferometer, micro-ring resonators, scandium-doped aluminum nitride, thin film.**

I. Introduction

ALUMINUM nitride (AlN) is generally assumed to be an attractive material for integrated photonics and optomechanims, and show potential applications in both micro electromechanical systems (MEMS) [1], [2], [3], [4] and integrated photonics [5], [6], [7], [8]. Among semiconductor materials, AlN is a complementary metal-oxide-semiconductor (CMOS) -compatible material with wide optical bandgap (6.2 eV) and broad transparency window reaching from the ultraviolet up to the mid-infrared wavelength regime [9], [10], [11]. Combined with the piezoelectric and second-order optical nonlinear effects, AlN is highly desired in integrated photonics. Studies have further shown that scandium (Sc) doping can increase the piezoelectric coefficient of Scandium-doped aluminum nitride (AlScN) while still maintaining CMOS compatibility [12], [13], [14]. Akiyama et al. [15] showed that the piezoelectric response of the $Al_{1-x}Sc_xN$ films gradually increased as the Sc concentration increased from $x = 0$ to $x = 0.43$, and the largest piezoelectric response of 27.6 pC/N was achieved in $Al_{0.57}Sc_{0.43}N$. Recently, Valerie Yoshioka et al. [16] reported that AlScN thin film possessed unusually large optical second-order susceptibility ($\chi^{(2)}$). For $Al_{0.64}Sc_{0.36}N$, the largest $\chi^{(2)}$ component d_{33} was enhanced to 62.3 $\pm$ 5.6 pm/V, which was at least one order of magnitude higher than AlN and twice the value of the extensively studied lithium niobate (LN). In addition, they measured the thin film propagation loss, with $Al_{0.64}Sc_{0.36}N$ exhibiting 17.2 dB/cm and $Al_{0.8}Sc_{0.2}N$ exhibiting 8.2 dB/cm [16]. Recent studies have further shown that AlScN is a ferroelectric semiconductor. Fengyan Y. et al. [17] demonstrated high-fidelity ferroelectric domain switching and periodic poling of Al-polar AlScN thin film.

Therefore, the AlScN photonic platform, which combines broad transparency window, significant piezoelectric effects, and large optical $\chi^{(2)}$, offers an expanded application range of integrated photonics. In 2020, Shiyang Zhu et al. [18] demonstrated the $Al_{0.85}Sc_{0.15}N$ photonic circuits on silicon (Si), showing a waveguide propagation loss of 9 $\pm$ 2 dB/cm and an micro-ring resonators (MRR) with a quality (Q) factor of 1.4×10^4. With increasing Sc concentration, loss increases as well, which remains to be a challenge. In addition, since the Sc-N bond energy (13.35 eV) is greater than the Al-N bond energy (11.52 eV), Sc doping also causes a challenge in the etching process compared to AlN [19]. Hence, further studies are necessary to unlock the potential of AlScN photonic platform and achieve integrated photonic devices with better performance.

In this work, we demonstrate the CMOS-compatible AlScN platform for integrated photonics based on 8-inch Si wafer. The AlScN thin films were prepared by reactive magnetron sputtering with 9.6% and 20% Sc concentrations. Results show that high-quality AlScN films with preferred c-axis orientation and low surface roughness have been deposited. Subsequently, we fabricated essential photonic devices based on the AlScN platform, including directional coupler (DC), Mach-Zehnder interferometer (MZI), and MRR. The extinction ratios (ERs) are more than 22 dB for $Al_{0.904}Sc_{0.096}N$ MZI and 24 dB

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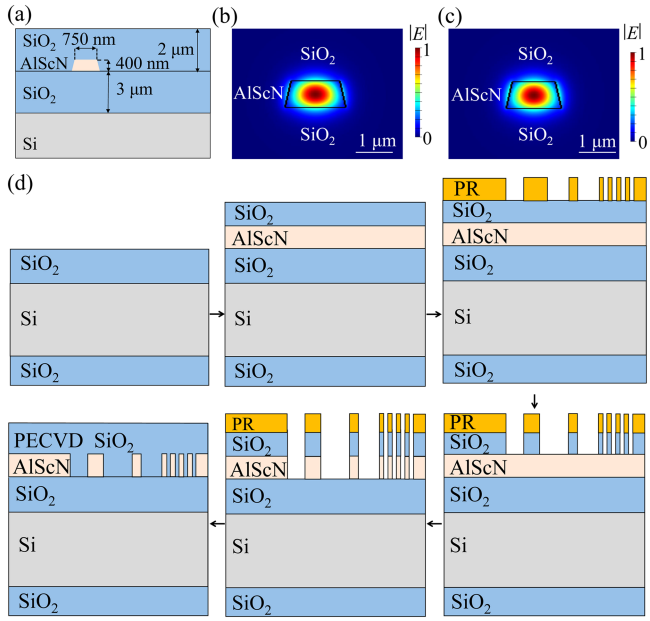


Fig. 1. (a) Cross-section schematic of the AlScN waveguide. (b)–(c) The electric field distribution of $Al_{0.904}Sc_{0.096}N$ and $Al_{0.8}Sc_{0.2}N$ single-mode waveguides, respectively. (d) Fabrication process.

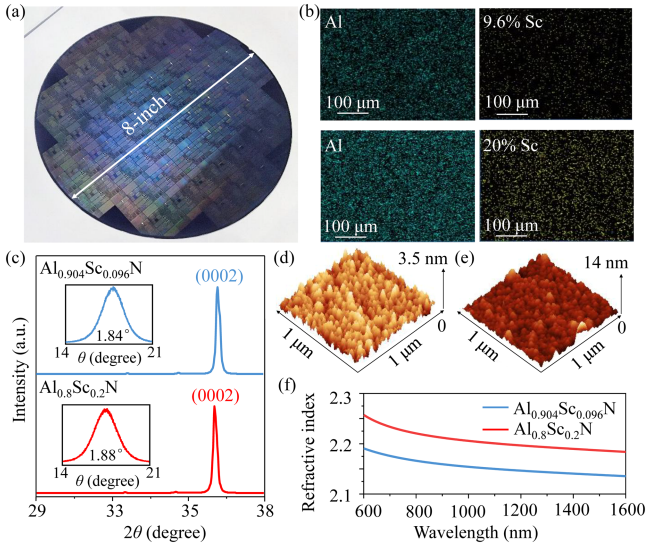


Fig. 2. (a) Image of the 8-inch AlScN wafer. (b) The elements distribution of the $Al_{0.904}Sc_{0.096}N$ and $Al_{0.8}Sc_{0.2}N$ thin films, respectively. (c) XRD 2θ pattern of the AlScN films and XRD rocking curve. (d)–(e) AFM images of the $Al_{0.904}Sc_{0.096}N$ and $Al_{0.8}Sc_{0.2}N$ thin films, respectively. (f) Measured refractive indexes as the function of optical wavelength.

for $Al_{0.8}Sc_{0.2}N$ MZI. The maximum Q factor of 3.2×10^4 for $Al_{0.904}Sc_{0.096}N$ MRR at 1550 nm is achieved, accompanied by a 100-μm radius bending waveguide propagation loss of 6.1 dB/cm. Meanwhile, for $Al_{0.8}Sc_{0.2}N$ MRR, the largest Q factor of 2.2×10^4 is achieved at 1550 nm wavelength, with a 100-μm R bending loss of 10.1 dB/cm. Benefiting from the broad transparency window, piezoelectric effect and second-order optical nonlinear effect, these basic components are expected to realize complex integrated optical circuits and brings potential for high efficiency and second-order harmonic generation, frequency comb generation, ferroelectric storage, and acousto-optic modulation.

II. Fabrication and Characterization

A. Design and Fabrication

Fig. 1(a) illustrates a cross-section schematic of the AlScN waveguide, wherein the AlScN film is positioned between two silicon dioxide (SiO_2) layers. The refractive indexes utilized in the simulation are extracted from the previous work [20], [21]. For $Al_{0.904}Sc_{0.096}N$, the refractive indexes is 2.07 at 1550 nm, while for $Al_{0.8}Sc_{0.2}N$, is 2.12 at 1550 nm, respectively. Fig. 1(b) and (c) show the simulated electric field distribution of the fundamental transverse electric (TE) modes near 1550 nm of single-mode $Al_{0.904}Sc_{0.096}N$ waveguide with a width of 850 nm and $Al_{0.8}Sc_{0.2}N$ waveguide with a width of 750 nm, respectively, and the sidewall angle is set to 75°.

The fabrication process for the devices is illustrated in Fig. 1(d). Firstly, 3-μm thermal SiO_2 was deposited on the 8-inch Si wafer. Then, 400-nm-thick AlScN thin film was prepared by reactive magnetron sputtering (Sigma Deposition System from SPTS). The AlScN thin films were deposited by Al-Sc target with a chemical composition of $Al_{1-x}Sc_x$ ($x = 0.096$ and 0.2) under a high purity nitrogen and argon mixture at 300 °C. Then plasma enhanced chemical vapor deposition (PECVD) was employed to deposit 500 nm SiO_2 as an etching hard mask. Photolithography was utilized for pattern definition. Inductively coupled plasma reactive ion etching (ICP-RIE) trials were carried out in the Omega etch system from SPTS. Cl_2/Ar etching scheme was utilized to transfer the patterns to the AlScN layer, using photoresist (PR) and SiO_2 as masks. After cleaning, the upper cladding of 2-μm SiO_2 was grown by PECVD, followed by smoothing the upper surface through a chemical-mechanical polishing (CMP) process.

B. Characterization of Films

Fig. 2(a) is the image of the 8-inch wafer. The film composition was measured by energy dispersive X-ray spectroscopy (EDX) with 20 kV acceleration voltage. The EDX element distribution diagrams in Fig. 2(b) illustrate the uniform distribution of Sc elements in the film without accumulation, with Sc measured percentages of 9 at.% and 18 at.%, respectively. The crystal orientation was investigated by X-ray diffraction (XRD). The peak intensities of the X-ray 2θ pattern appear near 36° as shown in Fig. 2(c), indicating that the films are deposited along the preferred crystal orientation (0002). As shown in the insets of Fig. 2(c), the full widths at half maximum (FWHM) of $Al_{0.904}Sc_{0.096}N$ and $Al_{0.8}Sc_{0.2}N$ X-ray rocking curves are measured to be 1.84° and 1.88°, respectively. As shown in Fig. 2(d) and (e), the surface root-mean-square (RMS) roughness values measured by atomic force microscopy (AFM) are 0.487 nm for $Al_{0.904}Sc_{0.096}N$ and 1.036 nm for $Al_{0.8}Sc_{0.2}N$, respectively. Additionally, the spectral ellipsometry test was also conducted on the deposited AlScN films. The measured

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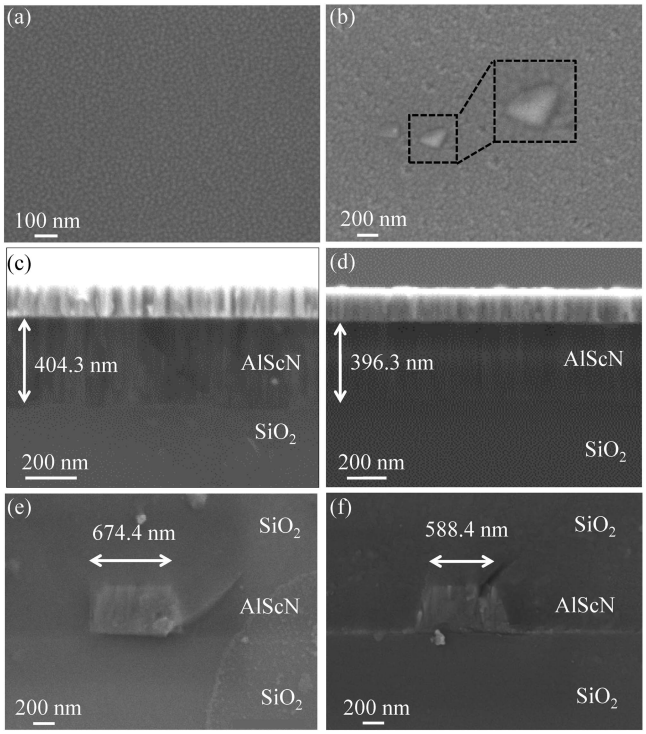


Fig. 3. (a)–(b) Surface morphology of $Al_{0.904}Sc_{0.096}N$ and $Al_{0.8}Sc_{0.2}N$ thin films, respectively. (c)–(d) Cross-sectional SEM images of $Al_{0.904}Sc_{0.096}N$ and $Al_{0.8}Sc_{0.2}N$ thin films, respectively. (e)–(f) Cross-sectional SEM images of $Al_{0.904}Sc_{0.096}N$ and $Al_{0.8}Sc_{0.2}N$ waveguides, respectively.

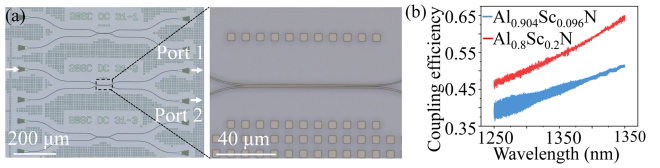


Fig. 4. (a) Optical microscope images of the fabricated DCs. (b) Coupling efficiencies with varying wavelength.

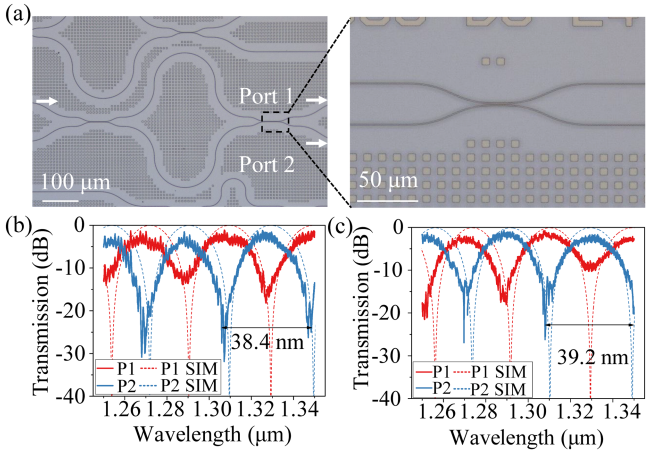


Fig. 5. (a) Optical microscope images of the fabricated MZI. (b) The MZI transmission spectra of $Al_{0.904}Sc_{0.096}N$ MZI. (c) The MZI transmission spectra of $Al_{0.8}Sc_{0.2}N$ MZI.

refractive indexes, depicted in Fig. 2(f), are 2.136 and 2.185 at the wavelength of 1550 nm for $Al_{0.904}Sc_{0.096}N$ and $Al_{0.8}Sc_{0.2}N$ films, respectively. These values are slightly higher than the values in our simulations.

Fig. 3 shows the scanning electron microscope (SEM) images of the AlScN films. The surface morphology evolution of thin films with increasing Sc content is shown in Fig. 3(a) and (b). When the concentration of Sc reaches 20%, noticeable abnormally oriented grains (AOGs) emerge on the surface [22], which shown in Fig. 3(b). The increase in AOGs leads to a deterioration in film quality and an increase in surface roughness, thereby impacting the performance of the devices. The $Al_{1-x}Sc_xN$ was deposited with a total thickness of 400 nm, and the thicknesses of the films were verified by cross-sectional SEM, as shown in Fig. 3(c) and (d). These images also reveal the well-aligned wide columnar grains in the growth direction, confirming highly orientation of AlScN thin film [23]. In order to contrast the relationship between Sc doping content and related results in the etching process, Fig. 3(e) and (f) show the SEM images of the same waveguide structure with 750-nm design width based on $Al_{0.904}Sc_{0.096}N$ and $Al_{0.8}Sc_{0.2}N$ thin films, respectively. The waveguides obtained through etching indicating a trapezoidal cross-section, with calculated sidewall angles of 77.5° and 73.5°, respectively. Due to the increased Sc concentration, the film quality and etching effect of $Al_{0.8}Sc_{0.2}N$ are slightly inferior to $Al_{0.904}Sc_{0.096}N$. With further optimization of the growth and fabrication process, both sidewall roughness and size bias are expected to be improved.

III. Devices Characterization

A. Directional Coupler

To efficiently guide on-chip optical signals, precise control over the interaction and polarization of evanescently coupled waveguide modes is essential. We fabricated several directional couplers, as illustrated in Fig. 4(a). Port 1 and Port 2 are the two output ports, and coupling efficiency is defined as the ratio of the optical powers at the Port 1 and Port 2. Fig. 4(b) shows the coupling efficiency of DCs from 1250 nm to 1350 nm. $Al_{0.904}Sc_{0.096}N$ DC consists of two waveguides with width = 700 nm, gap = 540 nm, and coupling length (L_c) = 21.7 μm. As for $Al_{0.8}Sc_{0.2}N$ DC, the waveguides width = 650 nm, gap = 540 nm, and L_c = 22.8 nm. It can be seen that the coupling efficiency is sensitive to wavelength. Moreover, due to the etched waveguide cross section being trapezoidal and the dimensions deviating from the design values, the distribution proportion of the optical mode field in the coupling gap between the waveguides is reduced, impacting the coupling between adjacent waveguides. These problems can be addressed through structural optimization, such as employing asymmetric DC [24] and bend DC [25].

B. Mach-Zehnder Interferometer

The MZI, a fundamental component in photonic integrated circuits, has broad applications for realizing complex optical components. The MZI we designed is shown in Fig. 5(a), which consisted of two DCs with a 50/50 splitting ratio and 540-nm

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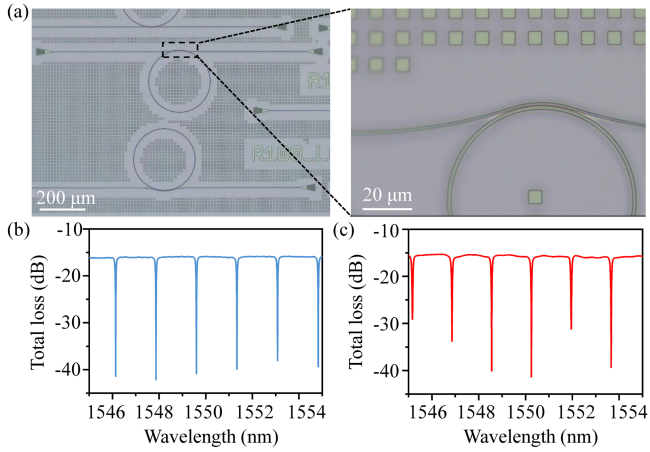


Fig. 6. (a) Optical microscope images of the fabricated MRR. (b) Measured TE mode transmission spectra of $Al_{0.904}Sc_{0.096}N$ MRR with radii of 100 μm. (c) Measured TE mode transmission spectra of $Al_{0.8}Sc_{0.2}N$ MRR with radii of 100 μm.

gap. Waveguides with width = 700 nm and length difference = 15.82 μm for $Al_{0.904}Sc_{0.096}N$, while width = 650 nm and length difference = 15.91 μm for $Al_{0.8}Sc_{0.2}N$, respectively. Output transmissions are recorded where Port 1 and Port 2 represent the two output ports, respectively. Fig. 5(b) and (c) show the transmission spectral lines normalized by the spectra of the grating couplers for $Al_{0.904}Sc_{0.096}N$ and $Al_{0.8}Sc_{0.2}N$ MZIs with the same designed free spectral range (FSR) of 38 nm. The dashed lines are the simulated spectral lines, and the shift of the spectral lines can be attributed to dimensional change of the the waveguide, resulting in smaller refractive indexes of the waveguide.

As shown in Fig. 5(b), the measured FSR is 38.4 nm for $Al_{0.904}Sc_{0.096}N$ MZI, which is slightly larger than the design value. This discrepancy is attributed to the reduction in the refractive indexes of the waveguide due to fabrication-induced dimensional deviation. As a result, the effective optical path difference of the delayed waveguide in MZI decreases, leading to an augmented FSR for the MZI. Fig. 5(c) shows $Al_{0.8}Sc_{0.2}N$ MZI transmission spectra. It can be seen that similar to $Al_{0.904}Sc_{0.096}N$, the FSR of the actual device is 39.2 nm. The ERs of MZIs extracted by the measured interference spectra were more than 22 dB for $Al_{0.904}Sc_{0.096}N$ MZI and 24 dB for $Al_{0.8}Sc_{0.2}N$ MZI. The smaller ER caused by Port 2 could be attributed to non-uniform optical power transmitted through the two arms of the MZI induced by DCs. According to interference theory, under the same conditions, the closer the amplitudes of the two arms are, the stronger the interference is and the greater the ER achieved [26]. So that the ER of MZIs can be improved by increasing uniformity of the splitting ratio.

C. Micro-Ring Resonator

MRR is one of the critical building blocks for highly compact photonic integrated circuits due to its wide application in switching and high-speed modulation. The MRR we designed is shown in Fig. 6(a). The $Al_{0.904}Sc_{0.096}N$ and $Al_{0.8}Sc_{0.2}N$ MRRs with 850-nm width and 750-nm width, respectively, 500-nm gap and 100-μm radius, show typical resonant properties, as illustrated in Fig. 6(b) and (c), respectively.

We analyzed the FSR, loaded Q factor, and intrinsic Q factor for the TE mode near the optical wavelength of 1550 nm. As for $Al_{0.904}Sc_{0.096}N$ MRR, the loaded Q factor is 3.2×10^4. The $Al_{0.8}Sc_{0.2}N$ MRR shows similar properties, and the maximum loaded Q factor is 2.2×10^4. The propagation loss can be estimated by the following equation [27], [28], [29]:

$$\text{Loss} = 10\log 10(e^{-\frac{2\pi n_g}{\lambda Q_i}}), \tag{1}$$

where n_g is the group index and it is obtained by the following equation [27], [28], [29]:

$$n_g = \frac{\lambda^2}{(\text{FSR})(L)}, \tag{2}$$

where L is the length of one round trip around the closed cavity. Q_i is the intrinsic Q factor of MRR. And it is obtained by the following equation [27], [28], [29]:

$$Q_i = \frac{2Q}{1 \pm \sqrt{T_0}}, \tag{3}$$

where T_0 is the normalized transmission power at the resonance wavelength. The intrinsic Q factors are 6.3×10^4 for $Al_{0.904}Sc_{0.096}N$ MRR and 3.9×10^4 for $Al_{0.8}Sc_{0.2}N$ MRR, respectively.

The waveguide bending losses are estimated to be 6.1 dB/cm for $Al_{0.904}Sc_{0.096}N$ MRR and 10.1 dB/cm for $Al_{0.8}Sc_{0.2}N$ MRR. As the increased concentration of Sc makes the dry etching more difficult, the final waveguide width is narrower and the sidewalls are rougher, resulting in larger waveguide propagation loss. The propagation loss of the waveguide is expected to be reduced by improving the fabrication process.

Furthermore, we have realized enhanced intrinsic Q factors by increasing the waveguide widths for both $Al_{0.904}Sc_{0.096}N$ and $Al_{0.8}Sc_{0.2}N$ MRRs to 1.1 μm and 1 μm, respectively, and reducing the radius to 35 μm. Fig. 7(a) and (b) shows the transmission spectra of $Al_{0.904}Sc_{0.096}N$ and $Al_{0.8}Sc_{0.2}N$ MRRs, and the FSRs of 5 nm and 4.9 nm are obtained from the measured spectrum, respectively. Fig. 7(c) and (d) depict the normalized spectra and the Lorentzian fittings of the fabricated MRRs [30], [31], [32], and the jitter of the data points can be attributed to the fluctuations of the grating couplers. The loaded Q factor, intrinsic Q and loss are 1.6×10^4, 1.1×10^5 and 3.5 dB/cm for $Al_{0.904}Sc_{0.096}N$, respectively. As for $Al_{0.8}Sc_{0.2}N$, the loaded Q factor, intrinsic Q factor and loss are 1.3×10^4, 4.6×10^4 and 8.5 dB/cm, respectively.

As shown in Table I, we compare the performance of AlScN MRRs near 1550 nm with reported works. The MRRs in this work exhibit higher Q factors compared to MRRs with similar Sc concentrations [33], [34]. The optical attenuation is largely increased with Sc doping compared to AlN [35] and increases with Sc concentration. The main reason is that the film quality will deteriorate and the waveguide sidewall roughness increases at higher Sc concentration. Nevertheless, it is expected to be effectively reduced by further optimization in the fabrication process. Firstly, to improve the quality of the films, aluminum

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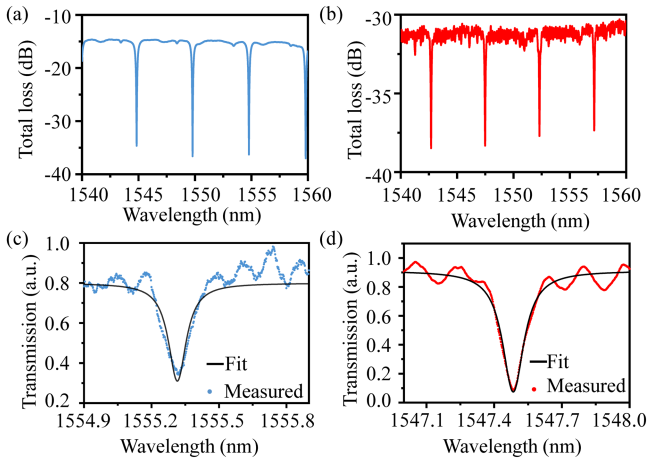


Fig. 7. (a) Measured TE mode transmission spectrum of $Al_{0.904}Sc_{0.096}N$ MRR. (b) Measured TE mode transmission spectrum of $Al_{0.8}Sc_{0.2}N$ MRR. (c) Measured $Al_{0.904}Sc_{0.096}N$ MRR TE mode resonance vally and the Lorentzian fitting. (d) $Al_{0.8}Sc_{0.2}N$ MRR TE mode resonance vally and the Lorentzian fitting.

TABLE I
COMPARISON OF PERFORMANCES FOR ALSCN MRRS

Material	Waveguide width (μm)	Propagation loss (dB/cm)	Q factor	Ref.
$Al_{0.9}Sc_{0.1}N$	1.2	6 (radius: 60 μm)	1.4×10^4	[33]
$Al_{0.85}Sc_{0.15}N$	N.A.	9 ± 2 (radius: 80 μm)	1.4×10^4	[18]
$Al_{0.8}Sc_{0.2}N$	2.5	N.A.	5.8×10^3	[34]
AlN	0.8	0.13 (radius: 60 μm)	2.8×10^6	[35]
$Al_{0.904}Sc_{0.096}N$	0.77	6.1 (radius: 100 μm)	3.2×10^4	this work
$Al_{0.904}Sc_{0.096}N$	1.1	3.5 (radius: 35 μm)	1.6×10^4	this work
$Al_{0.8}Sc_{0.2}N$	0.6	10.1 (radius: 100 μm)	2.2×10^4	this work
$Al_{0.8}Sc_{0.2}N$	1	8.5 (radius: 35 μm)	1.3×10^4	this work

oxide (Al_2O_3) seed layers can be used before AlScN film deposition. Compared with SiO_2, the lattice matching between c-oriented AlScN and Al_2O_3 is better [20]. Secondly, the BCl_3-based pretreatment can be utilized before the etching process to achieve smoother sidewall [36]. Furthermore, an annealing process can be applied for both film quality improvement and optical loss reduction [37], [38], [39].

IV. CONCLUSION

In conclusion, a CMOS-compatible photonic platform of AlScN was demonstrated through reactive magnetron sputtering with Sc concentrations of 9.6% and 20% on 8-inch Si wafers and on-chip photonic devices of DC, MZI, and MRR based on AlScN platform were fabricated and characterized. The ERs of MZIs are more than 22 dB for $Al_{0.904}Sc_{0.096}N$ and 24 dB for $Al_{0.8}Sc_{0.2}N$, respectively. As for MRR, the maximum loaded Q factor of 3.2×10^4 for $Al_{0.904}Sc_{0.096}N$ is achieved. Meanwhile, for $Al_{0.8}Sc_{0.2}N$, the largest loaded Q factor attained is 2.2×10^4. The waveguide bending losses are about 6.1 dB/cm for $Al_{0.904}Sc_{0.096}N$ MRR and about 10.1 dB/cm for $Al_{0.8}Sc_{0.2}N$ MRR. These results indicate that AlScN could offer a new platform for future integrated photonics.

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